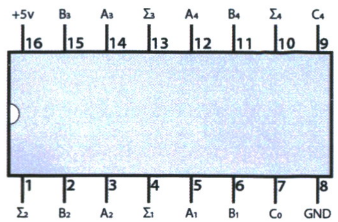
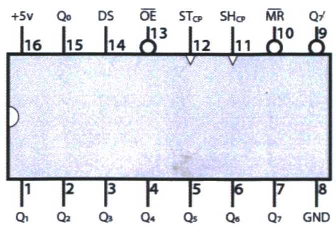
74LS283:
4-bit binary adder



This chip performs addition of two 4-bit binary numbers. $A+B=\Sigma$. Co and C4 are carry in and carry out for chaining multiple adders together

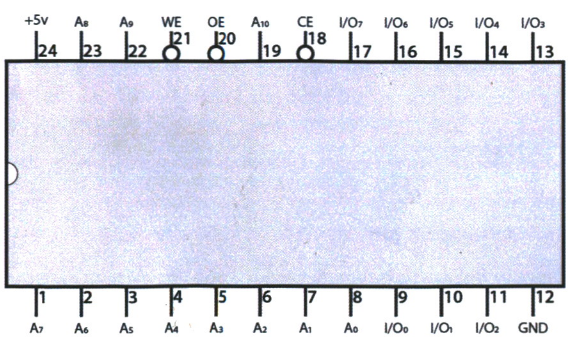
74HC595:
8-bit shift register



8-bit serial to parallel converter. Data is shifted in from DS on the positive edge of SH_{CP}. The shifted data is transferred to the output on the positive edge of ST_{CP}.

28C16: EEPROM

This EEPROM can be programmed with 2048 8-bit values. After applying an 11-bit address to the address lines, the stored value can be read from the 8 I/O lines.



If you get stuck

There's no denying it: Building a computer from scratch on breadboards is a big project. The best part is that by the time you're done, you'll have an incredible understanding of how the computer works. But at least some of that understanding will come because you had to track down some (possibly frustrating) problems. The good news is you're now part of a community of people working on this project and helping each other out. There are several good places to look for help:

- Comment threads on the YouTube videos.
- Comment threads at eater.net/8bit.
- Questions and discussions at reddit.com/r/beneater.

Caution:

You should never connect LEDs directly to a 5 volt source without a current limiting resistor. Doing so will damage the LED.

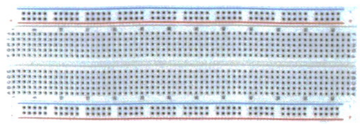
The outputs of most 74LS*xx* logic chips have internal resistors that let you to connect LEDs directly to the outputs without a resistor. (The 555 timer, 74189 RAM, 28C16 EEPROM, and 74HC595 *do not.*)

For the most reliable results, you may want to connect a 220Ω resistor in series with every LED anyway. I omitted many of these as a shortcut in the videos, but I've included enough 220Ω resistors in these kits if you choose (or need) to use them.

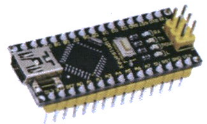
Components included in this kit

Note that some components might look slightly different than shown here or in the videos.

6 Breadboards



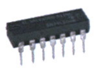
1 Arduino Nano



1 555 IC timer

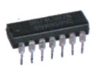
1 74LS138 3-8 line decoder

1 74LS00 Quad NAND gate



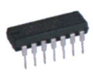
1 74LS139 2-4 line decoder

1 74LS02 Quad NOR gate



1 74LS161 4-bit counter

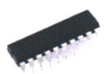
2 74LS04 Hex inverter



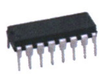
1 74LS173 4-bit D register

2 74LS08 Quad AND gate

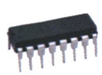
1 74LS273 8-bit D register



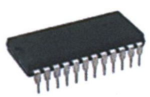
1 74LS107* Dual J-K flip-flop



2 74HC595 8-bit shift register



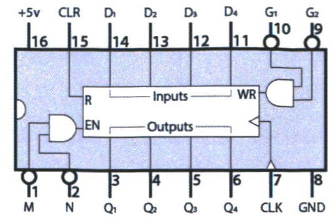
3 28C16 EEPROMs



*The 74LS107 replaces the 74LS76 dual JK flip-flop used in the videos. It's functionally the same, but ***be careful: the pinout is different!*** Pay attention to the function of each pin and make sure you hook it up correctly.

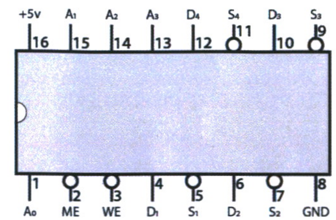
74LS173: 4-bit D register

This register stores 4 bits. You write data to it by holding G1 and G2 low and pulsing the clock.



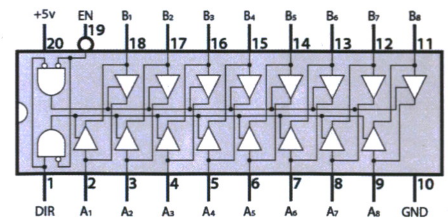
74189: 64-bit RAM

This static RAM holds 16 4-bit words. After applying a binary number to the four address inputs, data can be read or written to the memory.



74LS245: 8-bit bus transceiver

This allows 8- bit data from A to B or B to A depending on the DIR input. The EN input disables the device so that A and B are isolated.



74LS273: 8-bit D register

This register stores 8 bits. You write data to it by holding MR low and pulsing the clock.